

SAHAY-AI

What the System Will Do

National MVP capability document

Problem Statement **SIH26093** · AI-Based Real-Time Stress & Trauma Assessment

Team **ALT F4** · Ministry of Social Justice and Empowerment · NHAA 14566 and the Integrated Portal

Field	Value
Purpose	A complete statement of the capability the system delivers at the national MVP, for judges, reviewers and the team itself
Scope	Hindi, English and Hinglish end to end. Voice in Hindi and English. Speech emotion recognition in scope.
Covers	Victim journey · authority journey · system architecture · the SVI in full · what the system hears · safety architecture · privacy · differentiators
Tense	Written in the future tense throughout: this is the committed capability, not a status report
Companion documents	docs/research/local-research-analysis.pdf (evidence and projections) · docs/plan/ML_NATIONAL_MVP.md (milestones) · docs/contracts/CONTRACTS.md (frozen interfaces)
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1. The system in one page

SAHAY-AI is an AI-assisted intake, assessment and escalation module for the National Helpline Against Atrocities (14566) and the Integrated Portal. A person in distress speaks or types once, in their own language. The system conducts a bounded, consent-first intake, builds a structured case with linked evidence, prioritises it on a transparent index, and hands it to a trained helpline executive who decides every action.

Three sentences describe the whole design:

1. A deterministic state machine decides what the assistant says. A language model is permitted only to phrase that decision, and every sentence it produces is validated before it is spoken.
2. A synchronous crisis pre-check runs before anything else on every turn. When it matches, intake stops, a fixed approved script plays, the case becomes Critical, and a human takes over immediately.
3. Assessment is advisory. The executive decides, the decision is recorded separately from the AI proposal, and the person on the other end never receives a score, band, dimension or alert.

The shortest possible summary. It listens once, carefully, in the caller's own language; it never misses the call that cannot wait; and it hands a human being the evidence to act.

2. Scope of the national MVP

Eight capabilities define the delivered system. Each is a commitment, and each has an evidence path in Section~13.

#	Capability	What it means at the national MVP
T1	Languages	Hindi, English and Hinglish (code-mixed, in Roman or Devanagari script) end to end for text. Hindi and English for voice. Additional regional languages enter as text first, each gated by its own evaluation.
T2	Voice analysis	Speech is transcribed locally. Pauses, pitch variation, energy and speech rate are measured. Speech emotion recognition runs alongside a text-affect branch on the transcript. Dimension D4 is scored on voice sessions rather than abstaining.
T3	Text analysis	Every indicator the Problem Statement names is covered in every shipped language, with evidence links back to the utterances that produced it.
T4	Index and bands	A 0–100 Stress Vulnerability Index across nine dimensions, with Low, Moderate, High, Critical, and Needs Human Assessment as a fifth, first-class outcome.
T5	Recommendations	Counselling, legal aid, medical, police, witness protection and emergency support. Every recommendation stays awaiting_decision until a human acts on it.
T6	Voice output	Human-approved fixed scripts for the opening, closing, crisis and handoff states, with audio in every shipped language. Generated turns reach synthesis only after validation.
T7	Evidence	A per-language evaluation table: critical-event miss rate, detector precision and recall, word error rate and latency — each number carrying its evidence class.
T8	Safety invariants	All eight safety invariants hold, and each is enforced by an automated test that cannot be skipped.

3. The victim journey, end to end

The person reaching the helpline is the one bearing the cost of getting this wrong. Their journey is therefore designed around three guarantees: they tell it once, they are never assessed to their face, and a human is always one request away.

The path

STEP	WHAT THE PERSON DOES	WHAT THE SYSTEM DOES
1 ARRIVE	Calls 14566, opens the app, or uses the portal	Creates a session. Records the channel and the chosen language. No assessment has begun.
2 CONSENT	Chooses a language. Gives or declines consent.	S0 OPENING: a fixed, pre-recorded, human-approved script. "You are speaking with an AI. A human officer reviews everything. You can ask for a person at any time." Consent is stored as an artefact. DECLINE -> straight to a human queue. No scoring occurs at all.
3 SPEAK	Tells what happened, in their own words, uninterrupted	S1 FREE NARRATIVE: listens. Brief, neutral acknowledgement only. Never interrupts, never probes. Slots are extracted from what was said, not asked for.
4 ANSWER	Answers only what is still unknown. Eight to twelve turns in total.	S2 Are you safe right now? S3 Does anyone need medical help? S4 Who was involved, and when? S5 Are the threats continuing? S6 Is there someone with you? S7 Has a complaint been filed? S8 What help are you looking for? Any state already answered in the narrative is SKIPPED. >>> ON EVERY TURN, BEFORE ANY OF THE ABOVE <<< CRISIS PRE-CHECK -> match -> SX CRISIS INTERRUPT fixed approved script, no advice, no questions, band Critical, immediate human takeover, intake does not resume. >>> AVAILABLE IN EVERY STATE, ALWAYS VISIBLE <<< "Talk to a person" -> SH HUMAN HANDOFF assistant muted, executive claims the session, officer may write directly to the person.
5 CLOSE	Receives a reference number.	S9 CLOSING: a fixed script. Confirms the account is recorded, gives the reference, states what happens next. No promise of any outcome.
6 FOLLOW	Opens the timeline at any time.	Timeline shows the transcript, the status and the next step – and nothing else. The role filter strips every assessment field server-side.

What the person will never receive

On any channel, at any time, in any payload:

SVI score · risk band · any of the nine dimensions · emotion label
confidence value · safety alert · officer notes · recommendation status

This is enforced on the server by the role filter, never by the client, and an automated test asserts that a victim token cannot receive a dimension update or an assessment field. If the test fails, the build fails.

If the network drops

- The app captures the turn locally and queues it.
- On reconnect the session resumes at the same state, with the narrative intact.
- The person is never asked to start again, which is the single most damaging thing a distressed caller can be asked to do.

4. The authority journey, end to end

The executive console is the half of the product that decides whether any of this helps anyone. It is built on one principle: the system orders work, and a person disposes of it.

STEP	WHAT THE EXECUTIVE DOES	WHAT THE SYSTEM GUARANTEES
1 QUEUE	Reviews the live queue, ordered but complete	Ordered by band, confidence and waiting time. NEVER filtered. Nothing is hidden from a human because a model scored it low.
2 ALERT	Acknowledges any safety alert explicitly	An alert cannot be cleared by being ignored. Unacknowledged alerts escalate to a supervisor.
3 CLAIM	Claims a case	The case locks to one owner. Every action from here is attributed and timestamped.
4 READ	Opens the case packet	- full transcript and timeline - SVI band and the D1..D9 breakdown - the exact utterances behind every dimension - aggregate confidence - which overrides fired, and why - NEEDS HUMAN where the engine abstained - the gaps intake did not cover
5 VERIFY	Clicks a dimension	The source utterance highlights in the transcript. No score is ever shown without the words it came from.
6 DECIDE	Confirms / modifies / rejects / escalates	The AI proposal and the human decision are written to SEPARATE tables. A band override requires a written reason.
7 TAKE OVER	Optionally takes the live session	The assistant is muted for the rest of the session. The officer writes directly to the person. That text is human-authored: no AI generates, rephrases or sends it, and it never restarts intake.
8 DISPATCH	Routes to services	Counselling, legal aid, medical, police, witness protection, emergency. Each carries its rationale and evidence.
9 AUDIT	—	An append-only record of who saw what, who decided what, when, and on what basis.
10 SUPERVISE	Supervisor samples cases and reviews agreement	Confirmation rates are tracked. A rising rate of unchanged acceptance is treated as a SAFETY SIGNAL, not a success metric.

Why step 10 matters more than it looks. Automation bias — a human rubber-stamping a machine because the machine is usually right — is the characteristic failure of decision-support systems in high-pressure settings. Measuring it, and treating a high agreement rate as a warning rather than a win, is the difference between a system with a human in the loop and a system with a human in the way.

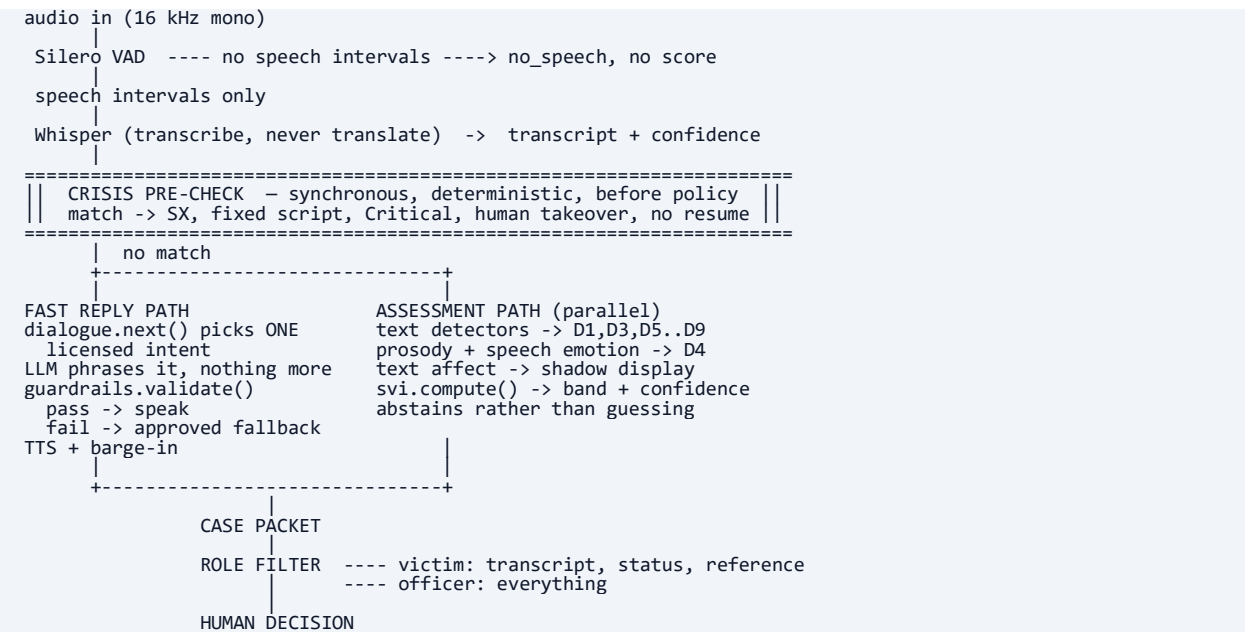
5. System architecture

The layers

CHANNELS	victim app (Android/iOS) . executive console Integrated Portal . existing chatbot . 14566 IVRS	
ADAPTERS	one normalised ingestion interface for every channel telephony . portal REST . CRM packet export . OIDC every adapter has a mock behind it	
EDGE	TLS . API gateway . rate limiting roles: victim . executive . supervisor . police . auditor	
SERVICES	Session Orchestrator Crisis pre-check Role-filtered fan-out	Dialogue state machine Guardrail validator Escalation and audit
AI LAYER	every provider behind an adapter, every one mockable VAD . ASR . prosody . speech emotion . text affect SVI engine (pure, deterministic, no I/O) LLM: phrasing only, and the system runs without it	
DATA	cases . timeline . evidence . audit (row-level) policy and scheme index . assessment queue audio and evidence objects, encrypted, auto-expiring AI proposals and human decisions in SEPARATE tables	
PLATFORM	containers on government cloud, an empanelled cloud, or an on-premises state data centre CPU-capable . offline-capable . India-resident	

One turn, from speech to reply

This is the path taken on every single turn of every session.



The property that makes this architecture unusual. The assessment path can be slow, can fail, or can be switched off entirely, and the conversation still completes correctly. The reply path never waits on it. The only synchronous intelligence in the turn is the crisis pre-check, and that is deterministic code rather than a model.

6. How the Stress Vulnerability Index works

The SVI is a prioritisation instrument, not a diagnosis. It exists to answer one operational question: of the cases waiting right now, which one should a human open first? Everything about its design follows from the requirement that an executive must be able to disagree with it in five seconds.

6.1 The nine dimensions and their weights

Dim	Meaning	Weight
D1	Immediate safety threat — can the person who caused harm reach them now	0.22
D2	Crisis and self-harm language	0.18
D3	Fear, intimidation and threats	0.13
D4	Acute distress — acoustic and emotional	0.12
D5	Trauma-associated indicators	0.08
D6	Social isolation, boycott, displacement	0.08
D7	Medical urgency	0.08
D8	Legal urgency	0.06
D9	Communication safety — whether the person can speak freely at all	0.05
	Total	1.00

The weights are published, not hidden, and they are labelled provisional in the product itself until an expert calibration study replaces them. A weight that cannot be shown to an executive is a weight that cannot be argued with, and a triage instrument nobody can argue with is not safe to deploy.

6.2 The arithmetic

Each dimension is scored 0-100.

$$SVI = \text{sum over } i \text{ of } (\text{weight } i \times \text{score } i)$$

BANDS	0 - 29	Low
	30 - 54	Moderate
	55 - 74	High
	75 - 100	Critical

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plus a fifth outcome that is not a band at all:
                                NEEDS HUMAN ASSESSMENT – no score is produced
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6.3 Renormalisation when a dimension cannot exist

D4 is acoustic. On a typed channel there is no voice, so D4 is not low — it is structurally unavailable, which is a different thing. The weights of the available dimensions are rescaled so that a text session is never penalised for being text:

$$\text{normalised svi} = \frac{\text{weighted sum available}}{\text{sum of available weights}}$$

With only D4 absent, the denominator becomes 0.88.

Every assessment stores: the scoring version, which dimensions were available, which were not, and the normalisation factor applied.

6.4 Hard overrides beat the arithmetic

A weighted average is the wrong instrument for a danger signal, because averaging is exactly what buries it. Two rules therefore sit above the arithmetic:

- Confirmed immediate danger (D1) above threshold forces the band to Critical, whatever the weighted score says.
- Confirmed crisis or self-harm language (D2) above threshold forces Critical on the same basis.

When an override fires, the case records which one and why, so the executive sees that the band was forced rather than computed.

6.5 Abstention is an outcome, not a failure

The system produces no score at all, and returns Needs Human Assessment, when any of the following hold:

- Aggregate confidence falls below 0.45.
- Audio quality is poor — low signal-to-noise, clipping, or under 800 ms of speech.
- Language confidence is low, so the transcript itself cannot be trusted.
- Consent was declined, in which case scoring is suppressed entirely rather than merely hidden.

A system that always returns a number is not more useful than one that sometimes says it does not know. It is less useful, because the executive loses the ability to tell the two situations apart.

6.6 A worked example

A voice session. The caller describes an assault by a neighbour who has since come to the house twice. She is frightened, the community has stopped speaking to her, and no complaint has been filed.

Dim	Score	Weight	Contribution	
D1	70	0.22	15.40	neighbour, has returned twice
D2	0	0.18	0.00	no crisis language
D3	65	0.13	8.45	explicit fear of retaliation
D4	55	0.12	6.60	voice: long pauses, unsteady pitch
D5	40	0.08	3.20	sleeplessness described
D6	60	0.08	4.80	community has stopped speaking
D7	10	0.08	0.80	no medical need stated
D8	30	0.06	1.80	no complaint filed
D9	25	0.05	1.25	speaking freely, alone at home
WEIGHTED SVI			42.30	-> band would be MODERATE
BUT: D1 is CONFIRMED and above threshold				

FINAL BAND		CRITICAL		overrides_applied: [D1_confirmed]

This example is the entire argument for the override rule. A person whose attacker can reach her tonight scores 42 on a weighted average, because seven of the nine dimensions are genuinely moderate. Averaging worked correctly and produced the wrong answer. The override exists because the arithmetic is right about the distribution and wrong about the danger.

The same case typed rather than spoken: D4 becomes unavailable, the weighted sum drops to 35.70, the denominator becomes 0.88, and the normalised SVI is 40.6 — still Moderate, still overridden to Critical. The override does not depend on the channel.

6.7 How it is presented

The presentation rules are part of the design, not a UI preference:

- Never a bare number. The breakdown and the confidence are always adjacent to the score.
- Every dimension carries the identifiers of the turns it was derived from, and clicking one highlights the utterance.
- Weights are labelled in-product as provisional, pending expert calibration.
- Trajectory is shown where a session has several assessments, so an executive can see whether distress is rising.

7. What the system hears: voice, prosody and emotion

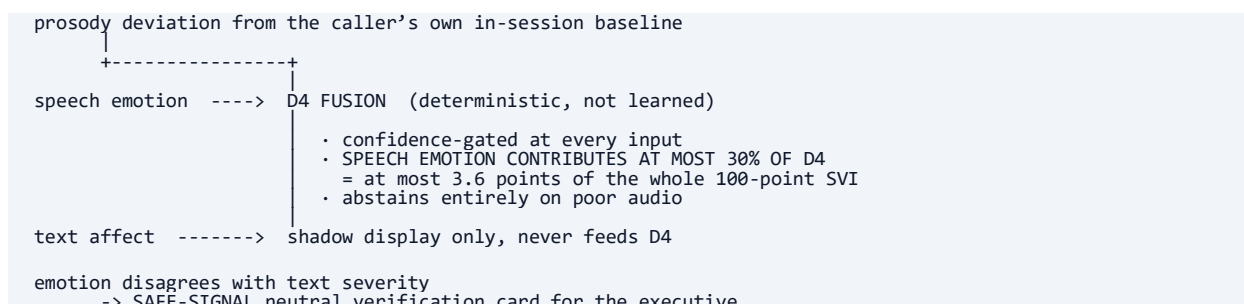
The Problem Statement asks for analysis of speech patterns, pauses, pitch variation and emotional indicators. The national MVP delivers this as three separate signals that converge on one dimension, under a strict contribution cap.

7.1 The three signals

Signal	What it measures	How it is produced
Prosody	Pitch mean and variability, energy, pause ratio, speech rate, response latency — measured against the caller's own baseline earlier in the same session, not against a population average	Deterministic feature extraction over the VAD speech intervals. Interpretable, and it alone answers the Problem Statement's "pauses and pitch variation".
Speech emotion	A five-class affect estimate over the utterance: neutral, happy, sad, angry, fearful	A speech emotion model selected on a predeclared rule, trained and evaluated on actor-disjoint splits, with a cross-corpus check.
Text affect	The same five classes, inferred from the transcript rather than the audio	A MuRIL head, chosen because MuRIL's pretraining includes transliterated Indian text — the Roman-script Hinglish callers actually use. Shown as a shadow signal.

"Happy" is deliberately kept as a class even though nobody is looking for happiness on this helpline. Without it, a model learns that loud, high-arousal speech means distress, and then mistakes animation for fear.

7.2 How they combine into D4



Why the 30% cap is the most important number in this section. Emotion recognition from voice is the least reliable component in the system and the easiest to over-trust. Capping it at 30 per cent of one dimension means it can move the total index by at most 3.6 points out of 100. It can inform a human. It can never, by itself, move a case between bands.

7.3 The rules that bind it

- Speech emotion and text affect never feed the crisis pre-check, routing, hard overrides or a forced band.
- The victim client never receives affect output of any kind.
- On the console it appears only inside the D4 breakdown, labelled as an auxiliary, uncalibrated signal.
- It never names a feeling as a fact about the caller, and it never names a condition.
- Poor audio, under 800 ms of speech, or low transcription confidence makes it abstain.

7.4 The promotion gate

Speech emotion feeds D4 in the product only if it passes a gate that was written down before any model was trained:

- Unweighted average recall of at least 0.45 across the five classes on in-domain Hindi recordings, against a chance level of 0.20.
- The same threshold met on in-domain English recordings.
- No single class with recall below 0.20.
- A negative control: on neutral recordings, its contribution to D4 stays at or below Moderate.

If the gate is not met, the signal is displayed as a shadow result, D4 is scored from prosody alone, and the measured number is reported as it stands. The capability is not removed; its authority is.

8. The safety architecture

Eight invariants hold across the whole system. Each is enforced by an automated test that is never skipped and never deleted.

#	Invariant	How it is enforced
1	A deterministic state machine chooses the intent; a model may only phrase it	<code>dialogue.next()</code> is a pure function with no I/O. Every generated sentence passes <code>guardrails.validate()</code> before synthesis, and a pre-written approved fallback is used when validation fails. The full intake runs with the model disabled.
2	Crisis language is detected before normal dialogue policy	The crisis pre-check is synchronous, deterministic and runs ahead of policy on every turn. A match forces state SX, a fixed approved script, Critical priority and immediate human takeover. Intake never resumes automatically.
3	Victim clients never receive assessment data	Server-side role filter, with a test asserting that a victim token cannot receive a dimension update and that the timeline payload contains no score, band, dimension, alert or confidence.
4	A human decides every recommended action	AI proposals and human decisions are written to separate tables with separate payloads. Recommendations remain <code>awaiting_decision</code> until an executive acts.
5	Confirmed danger or crisis overrides the weighted score	Hard overrides are applied after the arithmetic and recorded in <code>overrides_applied</code> .
6	Low confidence returns no score	Abstention thresholds on aggregate confidence, audio quality and language confidence, each independently tested.
7	AI disclosure and the route to a human stay available throughout	The disclosure banner persists for the whole session, and the handoff state is reachable unconditionally from every other state.
8	No real victim data in development	All corpora are authored fictional scenarios. Privacy scans run over every dataset before it enters the repository.

9. Language coverage

Language is not a feature toggle. A language ships when it has been evaluated, and not before.

Language	Text	Voice	What gates it
Hindi	Full	Full	Detector fixtures, validator rules, reviewed fixed scripts, word error rate measured
English	Full	Full	The same four
Hinglish (Roman or Devanagari)	Full	Evaluated separately	Treated as its own language with its own fixtures, not as a dialect of either parent
Additional regional languages	Enters text-first	Follows	Its own lexicons, its own fixtures written and reviewed by a fluent human, its own evaluation

Hinglish is treated as a first-class language rather than an afterthought because it is what people in distress actually speak. The text encoder is chosen specifically for it: MuRIL's pretraining covers transliterated Indian text, which is the case a model trained only on Devanagari handles worst.

Scale-out uses open Indian speech models and corpora — AI4Bharat's recognition, voices and datasets — so that adding a language is an adapter swap plus an evaluation rather than a procurement, and so that no audio in any language leaves the deployment.

10. Privacy, consent and data protection

Control	What the system will do
Consent	Recorded as a verifiable artefact before any assessment begins. A declined consent routes straight to a human and suppresses scoring entirely — not hidden, not computed and withheld. Suppressed.
Disclosure	The person is told they are speaking with an AI, that a human reviews everything, and that they may ask for a person at any time. The banner persists for the whole session.
Data minimisation	Raw audio is retained only as long as a configured window allows, then deleted automatically.
Encryption	Encrypted in transit and at rest, with envelope-encrypted keys.
Residency	India-resident on government or empanelled infrastructure. No audio or transcript leaves the deployment, in any language.
Access control	Role-based, enforced server-side, with the victim role structurally unable to receive assessment fields.
Retention and erasure	Configurable retention, with erasure on request.
Audit	Append-only record of every access and every decision: who saw what, who decided what, when, and on what basis.
Training data	No model training on identifiable victim data. Development uses authored fictional scenarios only.
Legal basis	Designed against the Digital Personal Data Protection Act 2023, with a named data protection officer and a documented breach-notification path.

11. What makes this different — the differentiators in detail

Each of the following is stated as a claim, the mechanism that makes it true, and what a competing system would have to change in order to match it. The last column is the point: most of these are architectural, so they cannot be added late.

1. The language model cannot decide anything

The claim. No model chooses what the assistant says, what it asks, or what happens next. The whole intake runs correctly with the language model switched off.

The mechanism. A deterministic twelve-state machine selects exactly one licensed intent. The model receives that intent and is permitted only to phrase it. Every sentence it returns passes a validator with explicit prohibitions before synthesis; on failure an approved pre-written line is used instead. Each intent has a reviewed fallback in every shipped language.

What it would take to copy. A prompt-based system cannot retrofit this. The state machine, the licensed-question scheme, the validator rule set and the per-intent multilingual fallbacks all have to exist before the first conversation, and the fallbacks have to be human-reviewed.

2. Crisis detection is deterministic and runs first

The claim. Self-harm and immediate-danger language are caught before any dialogue logic executes, on every turn, without a model being consulted.

The mechanism. A synchronous lexical pre-check with negation and quotation handling runs ahead of policy. A match forces state SX, a fixed counsellor-reviewed script, Critical priority and immediate human takeover, and intake never resumes on its own.

What it would take to copy. Any design that detects crisis inside the assessment pipeline has already lost the turn — and any design that asks a model to notice it has made the most safety-critical decision in the system the least predictable one.

3. The index is transparent, evidence-linked and arguable

The claim. Every score can be traced to the words that produced it, and an executive can disagree with any part of it in seconds.

The mechanism. Nine named dimensions with published weights, each dimension carrying the identifiers of the turns it derived from. A presentation contract forbids ever showing a bare number, and requires breakdown and confidence adjacent to the score.

What it would take to copy. A learned end-to-end score cannot be decomposed after the fact. Evidence linkage has to be designed into the scoring path from the beginning.

4. Abstention is a first-class outcome

The claim. The system returns "Needs Human Assessment" and produces no score when it is not confident, when audio quality is poor, or when the transcript cannot be trusted.

The mechanism. Predeclared thresholds on aggregate confidence, audio quality and language confidence, each independently tested, with the abstention recorded and explained rather than silently defaulted.

What it would take to copy. Systems built to always return a value have no way to distinguish "low risk" from "unknown", which are the two situations an executive most needs to tell apart.

5. The person is never assessed to their face

The claim. No score, band, dimension, emotion label, confidence value or alert ever reaches a victim client, on any channel.

The mechanism. A server-side role filter, with an automated test asserting that a victim token cannot receive a dimension update and that the timeline payload contains none of those fields. The test cannot be skipped.

What it would take to copy. Client-side hiding is not the same guarantee and does not survive a network inspection. This has to be enforced where the payload is built.

6. Emotion recognition is capped so it can never drive a decision

The claim. Speech emotion contributes at most 30 per cent of one dimension — at most 3.6 points of the 100-point index. It can inform a human. It cannot move a case between bands on its own.

The mechanism. A deterministic fusion step, confidence-gated, with the cap enforced in code and the signal abstaining entirely on poor audio. Emotion never feeds the crisis check, routing, overrides or a forced band, and it is promoted into the product only after passing a predeclared gate on in-domain recordings.

What it would take to copy. This is the opposite of the usual instinct, which is to feature emotion recognition prominently because it demonstrates well. Capping the least reliable component is a design decision that has to be made against one's own interest.

7. AI proposals and human decisions are structurally separate

The claim. The record of what the system suggested and the record of what a person decided are different tables with different payloads, and they can never be confused for one another.

The mechanism. Separate persistence, separate API surfaces, mandatory written reason on a band override, and an append-only audit log that binds each decision to an officer and a timestamp.

What it would take to copy. Storing a decision as a mutable field on the AI record makes it impossible to prove afterwards who decided what. It is a schema choice, and schema choices are expensive to reverse.

8. The queue is ordered but never filtered

The claim. Nothing is hidden from a human because a model scored it low.

The mechanism. Ranking affects presentation order only. Every case remains visible and openable, and unacknowledged safety alerts escalate to a supervisor rather than ageing out.

What it would take to copy. A system that suppresses low-scored cases is efficient right up to the moment it is wrong, and it will never tell anyone that it was.

9. Hinglish is a first-class language

The claim. Code-mixed, Roman-script Hindi-English is treated as its own language with its own lexicons, fixtures, validator rules and evaluation — not as a degraded form of either parent.

The mechanism. A text encoder chosen specifically because its pretraining covers transliterated Indian text, plus script detection, per-language fixtures and a separately measured evaluation slice.

What it would take to copy. Systems built for Devanagari Hindi and English separately handle the mixture worst, and the mixture is what a distressed caller actually speaks.

10. Every number carries its evidence class

The claim. The system distinguishes between results measured on data it was tuned on, on data it has been exposed to, and on a frozen blind corpus — and says which is which.

The mechanism. A contamination ledger, a freeze-and-hash protocol, and a rule that the first run against a corpus exposes it permanently. Measurement infrastructure refuses to run until a corpus is frozen.

What it would take to copy. This is the discipline that makes every other number in the project believable, and it is the one no demonstration will ever force a team to build.

11. It attaches to the helpline without changing it

The claim. The existing helpline keeps its authority, its procedures and its staffing. If the module is switched off, the helpline works exactly as it does today.

The mechanism. Every channel terminates at one normalised ingestion interface, each adapter has a mock behind it, and the case packet writes into the department's own system rather than creating a parallel record.

What it would take to copy. Integration designed after the fact tends to require the institution to change around the software. Here the interface was the first thing built.

12. It runs offline, on modest hardware, inside India

The claim. No audio leaves the deployment, no external API sits in the critical path, and there is no per-request fee. It runs on a government cloud, an empanelled cloud, or a state's own servers, from the same container image.

The mechanism. Self-hosted speech and language models, CPU-capable inference, and no proprietary managed service anywhere in the design.

What it would take to copy. A design that routes victim audio through a commercial speech API cannot be made sovereign later. It has to be built this way from the start.

12. What the system will deliberately never do

Stating this precisely is part of the design, and it is the answer to the hardest question the project will be asked.

It will never	Because
Diagnose, or name a psychiatric condition	It detects linguistic and acoustic indicators. A diagnosis is a clinical act performed by a qualified person, and nothing in this system is qualified to perform it.
Take an action on its own	Every recommendation stays awaiting_decision until a trained executive acts. The system proposes; it does not dispatch.
Give legal, medical or procedural advice	The validator prohibits it explicitly in every language, and the prohibition is red-teamed.
Promise an outcome, a timeline, an arrest or compensation	No one in the intake path is in a position to promise any of those, so the system is forbidden from implying them.
Ask for graphic or forensic detail	The licensed questions do not permit it. An intake is a first contact, not an investigation.
Show a score to the person being assessed	Enforced server-side and asserted by test.
Ship a language it has not evaluated	Language coverage is gated per language on its own fixtures, scripts and measurements.
Claim a number it has not measured	Every figure carries its evidence class, and unmeasured means unmeasured.

13. How every claim gets measured

Each capability in this document resolves to a number, and each number has an owner and a method.

Claim	Measured as	On what
Critical events are not missed	Critical-event miss rate with its Wilson confidence interval, reported beside the false-escalation rate	A frozen, human-authored blind corpus with two-reviewer adjudication on every crisis sample
Detectors work in every shipped language	Precision, recall and F1 per detector per language	Per-language fixtures, reported separately and never averaged into a single national figure
Speech is transcribed accurately	Word and character error rate by language and channel condition, including telephone-quality audio	Evaluation audio plus team-recorded fictional scripted speech in quiet and noisy conditions
Emotion recognition is good enough to inform D4	Unweighted average recall across five classes, with a confusion matrix and a cross-corpus check	Actor-disjoint splits, with a gate evaluated once on in-domain recordings held back for that purpose
It is fast enough to hold a conversation	End-to-end turn latency, p50 and p95, measured on the demonstration hardware with a real microphone	Instrumented voice sessions, not per-stage estimates
The index agrees with expert judgement	Weighted agreement against expert triage raters, plus the executive override rate	An expert calibration study, which also replaces the provisional weights
Nothing leaks to the victim client	A binary assertion: the payload contains no assessment field	An automated test that runs on every build and cannot be skipped
Officers are not rubber-stamping	Rate of unchanged acceptance of AI proposals, tracked over time	Production monitoring, with a rising rate treated as a safety signal

Companion documents: docs/research/local-research-analysis.pdf for measured evidence and scale projections; docs/plan/ML_NATIONAL_MVP.md for the milestone plan; docs/contracts/CONTRACTS.md, docs/dialogue/STATES.md and docs/HANDOVER.md for the frozen interfaces and the full specification.